

Technical Requirements Document (TRD)

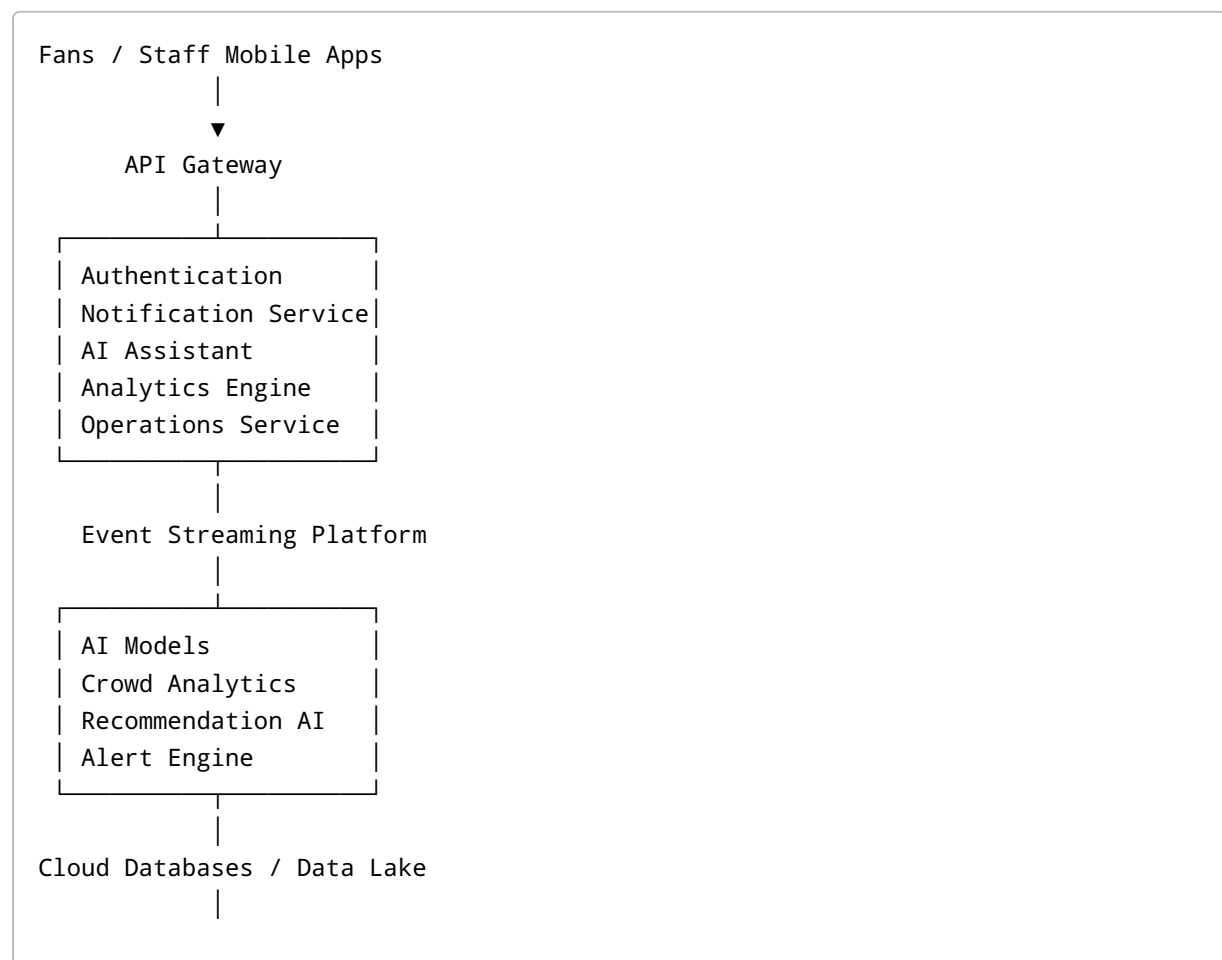
Product Name

StadiumAI – GenAI-Powered Stadium Operations & Fan Experience Platform

1. Technical Overview

StadiumAI is a cloud-native, AI-driven platform designed for real-time stadium operations. It integrates IoT devices, computer vision, edge computing, cloud services, and Generative AI to deliver intelligent assistance and operational insights during FIFA World Cup 2026.

2. High-Level System Architecture



3. Hardware Components

- Edge servers
 - GPU servers for AI inference
 - ESP32 microcontrollers
 - CCTV cameras
 - Environmental sensors
 - Occupancy sensors
 - Digital signage
 - Wi-Fi access points
 - BLE beacons
 - RFID/NFC ticket scanners
 - Emergency kiosks
 - Staff handheld devices
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4. Software Components

Backend

- Microservices architecture
- REST APIs
- GraphQL (optional)
- Event-driven services

Frontend

- Mobile applications
- Web dashboards
- Administrative portal

AI Components

- Large Language Model (LLM)
- Recommendation Engine
- Predictive Analytics
- Computer Vision
- Natural Language Processing
- Speech Recognition
- Translation Engine

Infrastructure

- Kubernetes
 - Docker
 - API Gateway
 - Message Queue (Kafka/RabbitMQ)
 - Redis Cache
 - Object Storage
 - Monitoring Platform
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5. Data Sources

Operational

- Ticketing systems
- Entry scanners
- Turnstiles
- Security systems
- Maintenance logs

IoT

- Temperature
- Air quality
- Occupancy
- Motion sensors
- Power consumption

Computer Vision

- CCTV
- Queue estimation
- Crowd density
- Incident detection

Fan Interaction

- Mobile app usage
- Navigation requests
- Purchases
- Surveys
- Feedback

External

- Weather services

- Public transport APIs
 - Emergency services
 - Match schedules
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6. AI Models

Generative AI

Purpose

- Conversational assistant
 - Fan guidance
 - Staff assistance
 - Knowledge retrieval
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Crowd Prediction Model

Purpose

- Predict congestion
 - Optimize movement
 - Suggest alternate routes
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Recommendation Engine

Purpose

- Food recommendations
 - Merchandise suggestions
 - Parking guidance
 - Personalized notifications
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Computer Vision Models

Purpose

- Crowd counting
- Queue estimation
- Security anomaly detection
- Restricted area monitoring

Predictive Analytics

Purpose

- Staff allocation
- Maintenance prediction
- Demand forecasting
- Incident probability estimation

7. Integration Points

Existing Stadium Systems

- Ticketing platform
- Access control
- CCTV
- Public Address (PA) system
- Emergency response systems
- Scoreboards
- Digital signage
- Parking management
- POS systems

Third-Party Services

- Payment gateways
- SMS providers
- Email providers
- Push notification services
- Mapping APIs
- Weather APIs
- Public transportation APIs

8. Scalability Requirements

The platform shall support:

- More than 120,000 concurrent users
- Horizontal scaling
- Auto-scaling Kubernetes clusters
- Distributed databases
- Load balancing

- Multi-region deployment
 - Elastic cloud resources
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9. Performance Requirements

API response time:

- Under 300 ms (average)

AI chatbot response:

- Under 2 seconds

Alert generation:

- Under 1 second

Dashboard updates:

- Every 2 seconds

Availability:

- 99.95% uptime
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10. Security Requirements

Authentication

- OAuth 2.0
- OpenID Connect
- Multi-Factor Authentication (MFA)

Authorization

- Role-Based Access Control (RBAC)
- Least privilege access

Encryption

- AES-256 encryption for data at rest
- TLS 1.3 for data in transit
- Secure key management

Monitoring

- Security Information and Event Management (SIEM)
 - Intrusion Detection System (IDS)
 - Intrusion Prevention System (IPS)
 - Continuous log monitoring
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11. ESP32 Security Strategy

ESP32 devices will serve as secure edge controllers and sensor gateways.

Device Security

- Secure Boot to verify trusted firmware
- Flash Encryption to protect stored firmware and sensitive data
- Hardware-based cryptographic acceleration
- eFuse key storage to prevent key extraction
- Device identity certificates for mutual authentication

Communication Security

- TLS 1.3 for encrypted communication
- Mutual TLS (mTLS) between ESP32 devices and servers
- Secure OTA (Over-the-Air) firmware updates with signed images
- Challenge-response authentication for connected devices

Data Protection

- Encrypt locally cached data using AES-256
- Minimize sensitive data stored on edge devices
- Rotate encryption keys periodically
- Validate sensor data integrity using HMAC/SHA-256

Threat Protection

- Firmware integrity verification
- Secure provisioning
- Replay attack protection with nonces and timestamps
- Device anomaly detection
- Automatic isolation of compromised edge devices

Note: ESP32 is a microcontroller platform rather than an encryption algorithm. The strategy above leverages its built-in hardware security features together with standard cryptographic algorithms (AES-256, TLS 1.3, SHA-256, HMAC) to protect data and resist hacking attempts.

12. Fault Tolerance

- Automatic failover
 - Service redundancy
 - Database replication
 - Retry mechanisms
 - Offline edge processing
 - Graceful degradation
 - Disaster recovery procedures
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13. Logging & Monitoring

Collect:

- API logs
- Application logs
- Security logs
- AI inference logs
- Infrastructure metrics
- Device health
- Network telemetry

Dashboards

- System health
 - Crowd analytics
 - Infrastructure utilization
 - Incident tracking
 - AI performance
 - Security monitoring
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14. Code Quality Standards

Development

- SOLID principles
- Clean Architecture
- Modular microservices
- Code reviews
- Static code analysis
- Consistent coding standards

Documentation

- API documentation (OpenAPI/Swagger)
 - Architecture diagrams
 - Sequence diagrams
 - Deployment documentation
 - Runbooks
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15. Testing Strategy

Testing Levels

- Unit Testing
 - Integration Testing
 - System Testing
 - End-to-End Testing
 - Performance Testing
 - Load Testing
 - Stress Testing
 - Security Testing
 - Penetration Testing
 - Accessibility Testing
 - Disaster Recovery Testing
 - User Acceptance Testing (UAT)
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16. Accessibility Requirements

The solution should align with **WCAG 2.2 Level AA** where practical.

Key requirements:

- Screen reader compatibility
 - Keyboard navigation
 - High-contrast support
 - Adjustable text sizes
 - Captioned multimedia
 - Multilingual interfaces
 - Voice interaction
 - Clear visual hierarchy
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17. Deployment Strategy

Development

→ CI/CD pipeline

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Testing

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Staging

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Production

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Continuous Monitoring

Deployment should support rolling updates, blue-green deployments, automated rollback, and infrastructure as code for reliable releases.

18. Technical Success Metrics

- 99.95% availability
 - Under 300 ms average API latency
 - Under 2-second AI response time
 - Zero critical security vulnerabilities in production
 - 100% encrypted communication between edge devices and cloud
 - Successful horizontal scaling under peak match-day traffic
 - Automated recovery from service failures within defined recovery objectives
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19. Technical Summary

StadiumAI combines cloud-native microservices, edge computing, IoT, computer vision, and Generative AI into a resilient platform capable of supporting the operational demands of FIFA World Cup 2026. By integrating securely with existing stadium infrastructure, employing industry-standard cryptography alongside ESP32 hardware security capabilities, and following rigorous engineering, testing, and

accessibility practices, the solution delivers a scalable, maintainable, and secure foundation for intelligent stadium operations and an enhanced fan experience.